

UTSUNOMIYA, Japan

WMO: 476150

Lat: 36.5498N

Lon: 139.8680E

Elev: 460

StdP: 14.45

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	24.5	26.4	7.8	8.3	40.3	10.6	9.6	40.0	24.6	40.6	21.1	42.2	5.0	10	0.426

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.8	92.8	78.0	90.4	77.1	87.8	76.0	79.5	88.9	78.4	87.0	77.4	85.3	9.2	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.1	143.7	84.0	76.0	138.3	82.9	75.0	133.7	82.0	43.5	88.9	42.3	87.5	41.2	85.7	83.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.7	17.3	14.5	DB	21.0	96.5	1.8	1.7	19.7	97.8	18.7	98.8	17.7	99.7	16.4	101.0	
			WB	19.0	81.2	1.8	1.3	17.7	82.1	16.7	82.9	15.7	83.6	14.4	84.5	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	58.5	37.4	39.3	45.6	55.5	64.7	70.8	78.0	80.1	73.2	62.7	51.3	41.8
	DBStd	15.59	3.82	4.75	5.85	6.26	5.18	4.89	5.33	4.77	5.73	5.35	5.50	4.90
	HDD50	1191	391	302	165	22	0	0	0	0	0	0	51	260
	HDD65	3787	855	720	601	288	69	9	0	0	5	108	412	720
	CDD50	4282	0	2	29	187	456	623	867	932	697	395	89	5
	CDD65	1405	0	0	0	3	59	182	403	467	253	38	0	0
	CDH74	10779	0	0	1	39	363	982	3355	4322	1592	125	0	0
	CDH80	3706	0	0	0	3	59	226	1270	1680	451	17	0	0
Wind	WSAvg	6.5	6.4	6.6	7.3	7.3	6.9	6.3	6.1	6.3	6.6	6.3	5.8	6.1
Precipitation	PrecAvg	64.2	1.7	1.9	3.3	5.0	6.1	7.0	8.7	8.5	10.0	6.6	3.5	1.9
	PrecMax	81.8	5.1	4.8	6.3	9.1	13.6	19.0	14.4	18.4	17.0	20.0	9.1	6.4
	PrecMin	44.3	0.2	0.2	0.7	1.1	2.8	2.2	2.7	2.3	1.1	0.5	0.0	0.2
	PrecStd	8.4	1.2	1.1	1.4	2.4	2.5	3.2	3.1	4.3	4.2	4.2	2.1	1.3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.1	62.2	70.3	78.3	85.3	89.4	95.2	95.7	91.3	82.1	69.7	61.0
		MCWB	45.6	50.5	53.3	59.2	65.6	73.5	78.3	79.3	76.6	69.3	58.2	51.2
	2%	DB	51.5	55.6	64.8	74.0	81.1	85.3	92.7	93.2	88.0	77.3	66.4	57.1
		MCWB	41.9	44.9	51.2	57.8	63.7	71.1	77.9	78.5	75.0	66.4	56.5	48.3
	5%	DB	48.8	52.1	60.8	70.5	78.0	82.4	90.0	91.1	85.1	74.1	63.8	54.1
		MCWB	39.4	41.9	49.1	56.9	62.8	69.4	77.0	77.8	73.8	64.2	55.0	46.0
	10%	DB	46.4	49.0	57.0	67.2	75.1	79.7	87.2	88.6	82.1	71.5	61.1	51.4
		MCWB	37.9	39.8	47.2	55.6	62.1	68.6	76.1	77.1	72.4	62.3	53.5	44.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.7	52.3	57.1	66.3	70.7	76.5	80.8	81.5	78.6	71.6	61.8	54.8
		MCDB	52.0	59.3	63.3	70.1	78.0	85.6	91.2	91.6	86.7	79.2	65.8	57.8
	2%	WB	43.8	46.7	54.2	62.0	67.6	74.4	79.2	80.1	77.0	68.9	59.4	50.6
		MCDB	49.2	53.1	61.1	69.8	75.4	81.4	88.8	89.6	83.9	74.3	63.5	54.8
	5%	WB	40.9	43.8	51.4	59.4	65.8	72.6	78.1	78.9	75.8	66.7	57.4	47.3
		MCDB	47.4	49.9	58.4	66.3	73.3	79.1	87.1	87.8	82.0	71.5	61.3	52.4
	10%	WB	38.7	41.1	48.3	57.2	64.1	70.7	77.0	77.9	74.5	64.7	55.2	45.0
		MCDB	44.9	47.7	55.6	63.6	71.4	76.9	85.2	86.0	80.3	69.8	59.4	50.3
Mean Daily Temperature Range	5% DB	MDBR	18.0	18.1	18.5	18.9	16.8	13.9	13.5	13.8	13.5	14.5	16.8	17.6
		MCDBR	21.0	22.3	23.9	23.7	21.9	18.5	17.8	17.2	16.4	17.1	19.0	19.9
		MCWBR	12.9	13.8	13.9	12.0	9.9	8.0	6.8	6.3	6.6	8.7	11.3	12.3
	5% WB	MCDBR	18.8	19.7	20.4	18.8	17.4	14.9	16.3	16.0	14.6	14.0	15.2	17.0
		MCWBR	12.8	13.5	14.1	11.6	9.6	7.2	6.6	6.3	6.6	8.3	10.5	11.9
Clear-Sky Solar Irradiance	taub	0.362	0.409	0.482	0.537	0.564	0.604	0.601	0.595	0.513	0.462	0.403	0.365	
	taud	2.237	2.093	1.925	1.851	1.838	1.807	1.870	1.876	2.051	2.114	2.207	2.266	
	Ebn at Noon	256	256	247	241	237	226	226	224	236	236	238	246	
	Edh at Noon	33	43	56	64	66	68	63	62	49	42	34	31	
All-Sky Solar Radiation	RadAvg	827	1033	1296	1510	1630	1476	1481	1467	1181	966	790	725	
	RadStd	45	77	72	129	163	115	231	162	96	91	67	45	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (36 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page